Sai Sri Teja Kuppa

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Education

Indian Institute of Technology, Madras **2023 - 2025**
MS in EE, Specialization in Computer Vision and AI. *CGPA: 8.00/10*

SASTRA Deemed University **2021 Batch**
Bachelor of Technology in Electronics and Communication Engineering *CGPA: 8.08/10*

Research Papers

ICCV Workshop (2025)	*FlareGS: 4D Flare Removal using Gaussian Splatting for Urban Scenes* Mayank Chandak, **K Sai Sri Teja**, Rahul Gopi Raju Matta, Vinayak Gupta, Kaushik Mitra. Paper — Webpage
ICCP (2025)	*PhotonSplat: 3D Scene Reconstruction and Colorization from SPAD* **K Sai Sri Teja**, Srividya V., Vinayak Gupta, Haejoon, Ashwin Shankaranarayana, Kaushik Mitra. Paper — Project Page — Code
ECCV Workshop (2024)	*DIVA – Deep Indic Virtual Apparel Try-On* **K Sai Sri Teja**, Hrishith Mitra, Girish Rongali, Kaushik Mitra. Paper — Dataset
CVPR Workshop (2024)	*MIPI 2024 Challenge on Nighttime Flare Removal: Methods and Results* Yuekun Dai, ..., **Kuppa Sai Sri Teja**, Jayakar Reddy A, Girish Rongali, Kaushik Mitra. Paper
Multimedia Tools and Applications (2022)	*Imaging-Based Cervical Cancer Diagnostics Using Small Object Detection-Generative Adversarial Networks* Elakkiya R., **K Sai Sri Teja**, Jegatha Deborah L., Bisogni C., Medaglia. Paper
FICTA – Springer (2022)	*3D CNN Based Emotion Recognition Using Facial Gestures* **K Sai Sri Teja**, Reddy T.V., Sashank M., Revathi A. Paper

Technical Focus Areas

Computer Vision	Image generation and Editing (diffusion), image restoration , 3D vision (Gaussian splitting), vision-language models (LLaMA 3.2, LLaVA), classification (ResNet, ViT), detection (YOLOv5, YOLOv12), novel cameras (SPAD, LiDAR)
Deep Learning	CNNs, transformers, attention mechanisms, encoder-decoder models, contrastive learning, self-supervised learning, quantization.
LLM & Agent Systems	Multi-agent orchestration (LangGraph), ReAct tool-calling agents, RAG (ChromaDB, vector retrieval), prompt engineering, MCP servers, session memory, LLM evaluation
Systems Engineering	Large-scale video/image processing, Model stress testing, model explainability, model training and pruning, research and module development

Experience

Immerso AI **Feb 2025 - Present**
Deep Learning Engineer / ML Engineer *Chennai, TamilNadu*

- **Project:** Text-to-Image (T2I) System – Eros *Feb 2025 – Nov 2025* **Overview:** Built large-scale dataset preparation and distributed training pipelines for Text-to-Image generation, enabling million-scale data processing, actor-specific personalization, automated captioning, and efficient model training and distillation.

  Key Contributions:

  * Designed automated pipelines for million-scale image datasets, including keyframe extraction, embedding-based redundancy removal, and large-scale storage workflows.
  * Implemented auto-tagging for Indic attributes and actor identification using embedding search and detection models.
  * Built scalable scraping and data ingestion systems for high-volume image collection.
  * Developed actor-tagging and dataset validation workflows using face embeddings and visual inspection tools.
  * Automated high-throughput caption generation using VLMs/LLMs for dataset enrichment.
  * Designed data pruning modules to improve dataset quality and training efficiency.
  * Trained actor-specific LoRA adapters for personalized generation.
  * Fine-tuned and trained MMDiT-based diffusion architectures using distributed training setups.
  * Developed production-grade Flow Matching and Diffusion training pipelines for high-fidelity image generation.

  Tools: Python, PyTorch, Diffusers, Ray, MosaicML, vLLM, FAISS, Grounding DINO, ArcFace, FiftyOne, Selenium, Scrapy, Bright Data.

- **Project:** AI Sprint Copilot – Eros *Dec 2025 – Feb 2026* **Overview:** Built and rolled out an org-wide AI sprint copilot at Eros by implementing LLM/agent research ideas incrementally (RAG, ReAct, multi-agent orchestration) into production modules. Teams across the organization use it to plan sprints from PRDs, track daily progress on GitHub and Plane, and onboard new members with a shared view of goals, blockers, and deliverables.

  Key Contributions:

  * Shipped capabilities paper-by-paper: retrieval-augmented Q&A over sprint docs, ReAct tool-calling over live repo/board data, and LangGraph supervisor routing with parallel summary + risk agents.
  * Deployed as the default sprint planning and tracking layer for engineering teams – PRD-to-ticket creation, daily standups, gap reports, code reviews, and slip/risk alerts used end-to-end across the sprint lifecycle.
  * Enabled faster onboarding for new hires: indexed PRDs, cited chat answers, and automated standup/report logs give newcomers sprint context without manual handoffs from leads.
  * Built a FastAPI platform with three chat modes (RAG, ReAct agent, supervisor), web UI, session memory, and FastMCP tools so teams and IDE agents query the same sprint state.
  * Integrated GitHub and Plane for real-time task sync, commit/diff analysis, contributor activity, timeline auto-push, and weekday automation via GitHub Actions.
  * Validated each module with pytest unit and integration tests before team rollout, covering RAG ingest/retrieval, agent routing, and end-to-end LLM flows.

  Tools: Python, FastAPI, LangChain, LangGraph, ChromaDB, Ollama, FastMCP, ReAct Agents, RAG, PyGithub, GitHub Actions, pytest, uvicorn.

- **Project:** World Models – Large Cultural Models (Video Generation) *Feb 2026 – Present* **Overview:** Contributor to the World Generation (video) pillar of Large Cultural Models – building temporally consistent video generation from unstructured real-world footage.

  Key Contributions:

  * Built camera pose estimation pipelines from unstructured videos to enable structured 3D-aware training data for world models.
  * Implemented person tracking and re-identification (ReID) pipelines for large-scale video dataset curation.
  * Contributed to video model training for world generation, targeting temporally consistent and culturally grounded scene synthesis.

  Tools: Python, PyTorch, COLMAP, OpenCV, YOLO, DeepSORT, Diffusers.

Detect Technologies Pvt Ltd — **July 2022 - Jan 2025**
Deep Learning Engineer - II — *Chennai, TamilNadu*

* **Project:** Auto Deep Learning Module
  Overview: Developed an automated system for building, training, and testing deep learning models without manual

intervention from engineers. The module streamlined data sampling, annotation, model training, and evaluation, significantly improving efficiency and scalability.

Activities performed:

- Built an automated deep learning pipeline covering data sampling, annotation (SAM, CLIP), pruning, training, evaluation, and explainability.
- Reduced manual annotation effort by 75% and improved workflow efficiency by 50% through intelligent data selection and automation.
- Optimized training with quantization and structured pruning, halving training time while maintaining performance.
- Scaled project throughput by 3x via modular and reproducible ML workflows.

Tools: Python, PyTorch, TensorFlow, Linux, Git, WandB, Docker, Captum.

* **Project:** Image Restoration for CCTV Photages

  Overview: Utilized neural networks to clean distorted images by addressing various challenging degradations such as motion blur, low-light, fog, and flare effects. Employed diffusion models and GAN-based architectures to restore and enhance image quality for improved downstream task performance.

  Activities performed:

  - Applied diffusion models (AutoDir) and GANs (Restormer, Uformer) for effective image restoration and enhancement.
  - Tackled diverse degradation types including motion blur, low-light conditions, fog, and flare.

  Tools: PyTorch, AutoDir, Restormer, Uformer, Diffusers

* **Project:** RAG based Gen AI Application

  Overview: Designed and implemented a multi-modal Fishbone Analysis system to support Root Cause Analysis (RCA) and Correction and Protection Analysis (CPA). The system processed client-provided image and text data to deliver actionable insights with enhanced interpretability and accuracy.

  Activities performed:

  - Integrated state-of-the-art Vision-Language Models (LLama 3.2, LLama 3.1, Molmo, LLava) with Large Language Models for robust multi-modal understanding.
  - Improved analysis reliability and interpretability for industrial RCA and CPA workflows using Retrieval-Augmented Generation (RAG) techniques.

  Tools: LLama 3.2, Python, Hugging Face Transformers, Ollama, Huggingface Endpoints, Streamlit.

Healthcare Technology Innovation Center - IIT Madras **Feb 2021 - June 2022**
Project Associate *Chennai, TamilNadu*

* **Project:** Motion Planning for Robotics

  Overview: Designed and implemented a complete software package in Python for robot-assisted needle-based spine surgery. The system includes kinematics, path planning, singularity detection, and collision avoidance. Gained hands-on experience working with real-time robots including KUKA (industrial robot) and Han's Elfin Robot (collaborative robot).

  Activities performed:

  - Built motion planning modules addressing forward/inverse kinematics, collision checks, and trajectory optimization for surgical precision. Developed and deployed robotics algorithms using ROS (Robot Operating System) for real-time execution and control.
  - Integrated camera sensors such as the Intel RealSense camera for depth sensing and 3D scanning in both simulations and real-world scenarios. Conducted extensive simulation experiments using PyBullet and transferred them to physical robots.

  Tools: Python, ROS, PyBullet, Intel RealSense, KUKA, Han's Elfin Robot

Achievements

* One among the world top 5 Teams in KUKA Robotics Challenge 2021(Team Aroki)
* Top 5 in the MIPI FLare removal challenge 2024(CVPR 2024).

Certifications and Workshops

* Participated 3D Vision Summer school(3DVSS) at IIIT Banglore(2024).